Joshua Santiago

Construction Coordinator / AI Developer / IT at Jacobs, Life Sciences Americas, Puerto Rico

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SUMMARY

AI-native software engineer who builds and operates production software for regulated industries and multi-agent systems with Claude Code. Sole technical person in the Jacobs office: built the reporting platform the project runs on and is building its field app. Previously delivered validated MES apps for a medical device maker, HIPAA-compliant storage for a health plan, and CI/CD at Granicus.

SELECTED PROJECTS

Joshua-OS: Claude Code operating system · In production — Aug 2026 to Present

Claude Code, Markdown agents and skills, Bash, PowerShell, Python, GitHub Actions

- Wrote the SOP and 9 global rule files every project follows (plan, build, verify, ship, learn; TDD with 80% coverage, pass^k on critical paths).
- Assembled and operate 66 Claude Code subagents (many from plugins), each with its own model and tool allow-list, plus Playwright, Exa and Supabase MCP servers.
- Built worktree.sh, which gives each builder agent its own git worktree (27 parallel lanes merged into main), and a tested Claude Code status mod.
- Made 190 commits since Aug 19, 2026; reviewer agents verify the work before a task is ticked done.

Jacobs reporting and compliance platform · In production — 2026 to Present

Python 3.11, openpyxl, FastAPI, vanilla JavaScript, hand-written SVG charts

- Replaced hand-assembled reporting with a 2,409-line Python ETL that finds each Excel and Primavera P6 workbook by its content, maps columns by header name and feeds a 26-view dashboard that needs 0 installs.
- Ported a 9,000-line React and FastAPI workforce app into the static dashboard, consolidating five compliance and commercial trackers that cover over 100 workers and more than 30 credential types.
- Made every number traceable to a source cell; absent fields show as unavailable, never zero.
- Built keyboard-navigable dialogs with focus return, audited against WCAG AA.

JevTrade: pre-registered LLM evaluation · Paper / sandbox — Oct 2026, five-day sprint

Python 3.12, pytest, ruff, uv, Alpaca paper API, TypeSafe Jev API

- Tested whether a commercial typed-judgment model calls BTC direction better than trivial baselines, with the pass bar fixed in advance. The model failed: 47.3% hit rate (95% CI 46.4 to 48.3) vs 50.9% for always up, across 17,005 calls scored from cached replays with a 10,000-resample block bootstrap. Published the negative result.
- Built an honesty harness: last 6 months sealed, walk-forward with purge, deflated Sharpe, and AST tests that keep the lab out of order code. 1,275 tests passing; lab and scorer at 100% line and branch coverage, as recorded.

Jacobs field app · In progress (Stage 1 done, pre-pilot) — Sep 2026 to Present

Expo 57, React Native 0.86, TypeScript, Supabase Postgres, Jest, Playwright, GitHub Actions

- Built a bilingual TypeScript app for iOS, Android and web (9 screens); Spanish and English carry 173 keys each, and CI fails the build if they drift.
- Enforced row-level security on all 12 Supabase Postgres tables (25 policies) and MFA at the database (aal2).
- Built an importer CLI whose fidelity oracle recounts app data against the source: 23 of 23 checks matched.
- Set up CI on GitHub Actions (typecheck, lint, Jest, pgTAP, Playwright with axe, gitleaks; actions pinned to SHAs). Tests: 461 Jest, 316 database and 21 end-to-end, as recorded.

EXPERIENCE

Construction Coordinator / AI Developer / IT — Apr 2026 to Present

Jacobs, Life Sciences Americas, Construction Services · Guaynabo, Puerto Rico

- Design, build and extend the project's reporting and compliance platform (above) from requirements gathered with project leadership, project controls and HSE; the project team adopted it for its reporting.

- Build the office's dashboards, reports, workflows and field app as its only technical person, and fix IT.
- Coordinate construction delivery and HSE compliance for a large capital program: certifications, inductions, job hazard analyses and prequalification.

Application Developer and Validation Engineer Dec 2025 to Mar 2026

Validation Resource Group · medical device manufacturing client · Cayey, Puerto Rico

- Designed, developed and validated Tulip MES apps replacing paper shop-floor processes, inspection logs and electronic batch record workflows.
- Converted SOPs and validation protocols into step-driven apps with batch tracking, equipment data capture, input validation, conditional logic and audit trails.
- Supported system testing, User Acceptance Testing and change tracking in Jira ahead of production release.

Back End Developer I Jun 2025 to Aug 2025

Alpine Health Technology · Guaynabo, Puerto Rico

- Configured secure data storage and backups to meet HIPAA requirements for a regional health plan client.
- Optimized SQL queries across health plan applications.
- Maintained full stack apps and supported their CI/CD in Azure DevOps.

Software Engineer May 2024 to Dec 2024

Granicus · San Juan, Puerto Rico

- Implemented Azure DevOps and GitHub Actions CI/CD pipelines that test, integrate and deploy ASP.NET apps.
- Built and maintained full stack web apps across .NET and Java back ends for public sector education and justice case management systems (C#, VB.NET, Java, JavaScript, SQL Server).
- Wrote technical, user and API documentation, and delivered features with design and product teams.

Software Engineer Intern Oct 2023 to Apr 2024

Granicus · San Juan, Puerto Rico

- Improved CI/CD deployment quality through code review and automated testing.
- Supported full stack maintenance in C#, SQL and JavaScript.

SKILLS

Agents & tooling: Claude Code (primary development environment), subagents and skills, MCP servers (Playwright, Exa, Supabase), tool allow-lists, LLM evaluation (pre-registration, bootstrap CIs, cached replays), Firecrawl

Languages & frameworks: Python, TypeScript, JavaScript, Node.js, C# and .NET, SQL, Java (read and debug), C, FastAPI, pydantic, React, React Native and Expo, Django, HTML and CSS

Data & infra: PostgreSQL and Supabase (row-level security), Microsoft SQL Server, SQLite, ETL with openpyxl, Azure (AZ-204), AWS, Docker, Git, GitHub Actions, Azure DevOps, Linux

Practices: test-driven development (pytest, Jest, pgTAP, Playwright with axe), CI/CD with secret scanning (gitleaks), MES and electronic batch records, validation protocols, audit trails, UAT, HIPAA data handling, Agile, Jira

EDUCATION

Bachelor of Science, triple major in Computer Information Systems, Accounting, and Business Administration 2017 to 2022

Peru State College

Software Engineering, 9 Month Certificate Program Jun 2023 to Mar 2024

Holberton Coding School, San Juan, Puerto Rico · Capstone: HealHub, a health app (Node.js, Python, Django, PostgreSQL, AWS)

CERTIFICATIONS

- Microsoft Certified: Azure Developer Associate (AZ-204), June 2024
- Tulip Advanced Developer Certification, Dec 2025 · Tulip Basic Developer Certification, Dec 2025
- Cybersecurity Certification (Holberton), Mar 2024

LANGUAGES

Spanish: native · English: fluent